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Academic Position

Postdoctoral Fellow, Kavli IPMU, University of Tokyo
Oct. 2023 - Present (Funded until Oct. 2026)

Education and Degrees

Ph.D., Department of Physics and Astronomy, School of Arts and Sciences, University of Pennsylvania
Advisor: Prof. Mirjam Cvetič, Co-Advisor: Prof. Jonathan J. Heckman
Aug. 2018 - Aug. 2023

B.S. in Physics, School of Physics, Peking University
Major: Physics
Sep. 2014 - Jul. 2018

Publications

27. Jiakang Bao, Noppadol Mekareeya, Gabi Zafrir, and Hao Y. Zhang, Orbi-Instantons and Class S Theories of Type D, *arxiv*: 2602.03931

26. Motoo Suzuki, Ling-Xiao Xu, Hao Y. Zhang, Spurion Analysis for Non-Invertible Selection Rules from Near-Group Fusions, *arxiv*: 2508.14970

25. Saghar S. Hosseini, Yuji Tachikawa, Hao Y. Zhang, Type I Anomaly Cancellation Revisited, *arxiv*:2505.07933

24. Xingyang Yu, Hao Y. Zhang, von Neumann Algebra and non-Invertible Symmetries, *arxiv*:2504.05374, *SciPost Phys.* 19 (2025) 154

23. Jiakang Bao, Hao Y. Zhang, Atomic Higgsings of 6D SCFTs II: Induced flows, *arxiv*: 2501.03313, *JHEP* 05 (2025) 125
22. Mirjam Cvetič, Markus Dierigl, Ling Lin, Ethan Torres, and Hao Y. Zhang, Frozen Generalized Symmetries, *arxiv*:2410.07318, *Phys.Rev.D* 111 (2025) 2
21. Jiakang Bao, Hao Y. Zhang, Atomic Higgsings of 6D SCFTs, *arxiv*: 2409.17224, *JHEP* 05 (2025) 124
20. Hao Y. Zhang, K-theoretic Global Symmetry in String-Engineered QFT and T-duality, *arxiv*: 2404.16097, Submitted to *Journal of High Energy Physics*
19. Yuji Tachikawa, Hao Y. Zhang, On a $\mathbb{Z}_3$ -valued Discrete Topological Term in 10D Heterotic String Theories, *arxiv*: 2403.08861, *SciPost Phys.* 17 (2024) 077
18. Justin Kaidi, Yuji Tachikawa, Hao Y. Zhang, On a Class of Selection Rules without Group Actions in Field Theory and String Theory, *arxiv*: 2402.00105, Published in *SciPost Physics*, with inclusion of an extended abstract in *SciPost Selections*
17. Craig Lawrie, Xingyang Yu, Hao Y. Zhang, Intermediate Defect Groups, Polarization Pairs, and Non-invertible Duality Defects, *Phys.Rev.D* 109 (2024) 2, 026005
16. Jonathan J. Heckman, Max Hubner, Ethan Torres, Xingyang Yu, Hao Y. Zhang, Top Down Approach to Topological Duality Defects, *Phys.Rev.D* 108 (2023) 4, 046015
15. Jonathan J. Heckman, Max Hubner, Ethan Torres, Hao Y. Zhang, The Branes Behind Generalized Symmetry Operators, *Fortschr. Phys.* 2022, 2200180
14. Jonathan J. Heckman, Craig Lawrie, Ling Lin, Hao Y. Zhang and Gianluca Zoccarato, 6D SCFTs, Center-Flavor Symmetries, And Stiefel—Whitney Compactifications, *Phys.Rev.D* 106 (2022) 6, 066003
13. Mirjam Cvetič, Markus Dierigl, Ling Lin and Hao Y. Zhang, All Eight and Nine Dimensional String Vacua From Junctions, *Phys.Rev.D* 106 (2022) 2, 026007
12. Michele del Zotto, Jonathan J. Heckman, Shani Nadir Meynet, Robert Moscrop, and Hao Y. Zhang, Higher Symmetries of 5d Orbifold SCFTs, *Phys.Rev.D* 106 (2022) 4, 046010
11. Mirjam Cvetič, Markus Dierigl, Ling Lin and Hao Y. Zhang, Gauge Group Topology of 8D Chaudhuri-Hockney-Lykken Vacua, *Phys.Rev.D* 104 (2021) 8, 086018
10. Mirjam Cvetič, Markus Dierigl, Ling Lin and Hao Y. Zhang, Higher-Form Symmetries and Their Anomalies in M/F-Theory Duality, *Phys.Rev.D* 104 (2021) 12, 126019
9. Jonathan J. Heckman, Sandipan Kundu and Hao Y. Zhang, EFT of 6D SUSY RG Flows, *Phys.Rev.D* 104 (2021) 8, 085017.

8. Jonathan J. Heckman, Craig Lawrie, Thomas B. Rochais, Hao Y. Zhang and Gianluca Zoccarato, S-folds, String junctions, and 4D $N=2$ SCFTs, *Phys.Rev.D* 103 (2021) 8, 086013
7. Mirjam Cvetič, Markus Dierigl, Ling Lin and Hao Y. Zhang, String Universalities and Non-Simply-Connected Gauge Groups in 8d, *Phys.Rev.Lett.* 125 (2020) 21, 211602
6. Fabio Apruzzi, Marco Fazzi, Jonathan J. Heckman, Tom Rudelius and Hao Y. Zhang, General prescription for global $U(1)$'s in 6D SCFTs, *Phys.Rev.D* 101 (2020) 8, 086023
5. Falk Hassler, Jonathan J. Heckman, Thomas B. Rochais, Tom Rudelius and Hao Y. Zhang, T-branes, String Junctions, and 6D SCFTs, *Phys.Rev.D* 101 (2020) 8, 086018
4. Mirjam Cvetič, Ling Lin, Muyang Liu, Hao Y. Zhang and Gianluca Zoccarato, Yukawa Hierarchies in Global F-theory Models, *JHEP* 01 (2020) 037
3. Vijay Varma, Davide Gerosa, Leo C. Stein, François Herbert and Hao Zhang, High-accuracy mass, spin and recoil predictions of general black-hole merger remnants, *Phys.Rev.Lett.* 122 (2019) 1, 011101
2. Hao Zhang, Richard de Grijs, Chengyuan Li and Xiaohan Wu, No Evidence for Chemical Abundance Variation in Intermediate-age Cluster NGC 1783, *The Astrophysical Journal*, 853:186 (9pp), 2018 February 1
1. Hao Zhang, Daniel J. Eisenstein, Lehman H. Garrison and Douglas W. Ferrer, Testing the Detection Significance on the Large Scale Structure by a JWST Deep Field Survey, *The Astrophysical Journal*, 875 (2019) 2, 132

Research Talks

Green-Schwarz Anomaly Cancellation in Type I String Theory via Differential KO Theory (25+5 min), 7th Peng Huanwu Forum for Young Scientists in Physics and Relevant Interdisciplinary Studies, ITP-CAS, Beijing, Feb. 2026 (Best Presentation Award)

Non-Invertible Selection Rules in QFT and String Theory (1h), Seminar at ITP-CAS, Dec. 2025

Type I Anomaly Cancellation and Differential KO Theory (1h15min), String Theory, Supergravity, and Related Topics, USTC, Hefei, Anhui, Aug. 2025.

von Neumann Subfactors and Gauging Non-invertible Symmetries (1h), Generalized symmetries in HEP and CMP, Peking University, Beijing, Jul. 2025

von Neumann Subfactors and Gauging Non-invertible Symmetries (1.5h), Seminar at KITS-UCAS, Jul. 2025

Type I Anomaly Cancellation and Differential KO Theory (1h30min), Seminar at School of Physics, Peking University, Jun. 2025

von Neumann Subfactors and Gauging Non-invertible Symmetries (1.5h), String Lunch Seminars at Kavli IPMU, May 2025

Z_3 -valued Theta Angle in 10D Heterotic Strings and Topological Modular Forms, (25+5 min), 6th Peng Huanwu Forum for Young Scientists in Physics and Relevant Interdisciplinary Studies, ITP-CAS, Beijing, Jan. 2025 (Best Presentation Award)

Atomic Higgsings of 6D SCFTs, (online 1-hour) Quiver Meeting organized by Dr. Julius Grimminger at Oxford University, Jan. 2025

Atomic Higgsings of 6D SCFTs, 1-min Gong Show and Poster Presentation, 19th Asian Winter School at Sanya China, Jan. 2025 (*Third Prize of Best Poster Award*)

Atomic Higgsings of 6D SCFTs, (online 40 + 5 min) Young Scientist Forum held by Yau Center of Southeastern University, Dec. 2024

Spacetime Avatar of Non-invertible Symmetries on the String Worldsheet, (online 1-hour) Quantum Spacetime Seminar Series, Tata Institute of Fundamental Research, Oct. 2024

Atomic Higgsing of 6D SCFTs via Geometric Engineering (1.5 hour), Symmetry and VOA workshop, Atami, Shizuoka, Japan, Oct. 2024

Z_3 -valued Theta Angle in 10D Heterotic Strings and Topological Modular Forms (1 hour), Friday Lunch Seminar, Department of Physics and Astronomy, University of Pennsylvania, Sep. 2024

K-theoretic Global Symmetries and T-duality (1 hour), Rutgers NHETC Seminar, Sep. 2024

Z_3 -valued Theta Angle in 10D Heterotic Strings (20 minutes), IHES Summer School on Symmetries and Anomalies, Jul. 2024

Spacetime Avatar of Non-invertible Symmetries on the String Worldsheet (1 hour), Department of Physics, Osaka University (Toyonaka Campus), Jun. 2024

K-theoretic Global Symmetries and T-duality (1 hour), online Symmetry Seminars at Mathematical Institute, Oxford University, May 2024

Z_3 -valued Theta Angle in 10D Heterotic Strings and Topological Modular Forms (1.5 hour), String Lunch Seminars at Kavli IPMU, May 2024

Spacetime Avatar of Non-invertible Symmetries on the String Worldsheet (1 hour), Department of Physics, University of Tokyo (Hongo Campus), May 2024

Z_3 -valued Theta Angle in 10D Heterotic Strings and Topological Modular Forms (1.5 hour), BIMSA String Theory Group, Apr. 2024

Spacetime Avatar of Non-invertible Symmetries on the String Worldsheet (1 hour), Yau Mathematical Sciences Center, Tsinghua University, Apr. 2024

Z_3 -valued Theta Angle in 10D Heterotic Strings and Topological Modular Forms (1 hour), KEK Tsukuba Theory Group, Apr. 2024

Non-Invertible Duality Symmetry in 6D and "Polarization Pair" (1 hour), Theoretical High-Energy Theory Seminars, Department of Physics and Astronomy, University of Pennsylvania, Dec. 2023

Non-Invertible Duality Symmetry in 6D and "Polarization Pair" (1 hour), Simons Center of Geometry and Physics, Dec. 2023

Non-Invertible Duality Symmetry in 6D and "Polarization Pair" (1 hour), Swampland Seminar Series, Department of Physics, Harvard University, Nov. 2023

Discrete Aspects of String Landscape and String Universality in High Dimensions (1 hour), Department of Math, Tsinghua University, Sep. 2023

Non-Invertible Duality Symmetry via Relative QFT and "Polarization Pair" (1 hour), 1-hour in-person seminar, School of Physics, Peking University, Sep. 2023

Discrete Aspects of String Landscape and String Universality in High Dimensions (1 hour), Texas A&M University, Apr 2023

Discrete Aspects of String Landscape and String Universality in High Dimensions (1.5 hour), Penn string theory group, Apr 2023

Generalized Symmetry Operators from Branes, 30 mins online, Seminar Series in String Phenomenology, Oct 2022

Generalized Symmetry Operators from Branes, 5-min gongxi and poster presentation, Generalized Global Symmetries, Quantum Field Theory, and Geometry, SCGP, Stony Brook

String universality in 8D and 9D, Graduate Hep-th Seminar, University of Pennsylvania, May 2022

Higher Symmetries of 5d Orbifold SCFTs, 30 mins in-person talk, Geometrization of (S)QFT in $D \leq 6$, Aspen, Feb 2022

Effective Field Theory of 6D (1,0) Renormalization Group Flows, Online Poster Session, Strings 2021

String Universality and Non-Simply-Connected Gauge Groups in 8D, 30 mins online talk, on Seminar Series in String Phenomenology, Sep 2020

General prescription for global $U(1)$'s in 6D SCFTs, Online gong show + poster presentation, on Quantum Field Theory and Geometry Summer School, Jun 2020

Compactification of Heterotic Strings, Graduate String Seminar, May 2019

Keck Spectroscopy of NGVS Sources: Milky Way Halo Star Kinematics, In-person poster presentation, on the 229th American Astronomical Society (AAS) Meeting, Jan 2017

Awards

2026 Best Presentation Award, 7th Peng Huanwu Forum for Young Scientists in Physics and Relevant Interdisciplinary Studies, Institute of Theoretical Physics-Chinese Academy of Sciences

2025 Best Presentation Award, 6th Peng Huanwu Forum for Young Scientists in Physics and Relevant Interdisciplinary Studies, Institute of Theoretical Physics-Chinese Academy of Sciences

2025 Best Poster Award (Third Prize), 19th Asian Winter School, Sanya

2019 Orville Phillips Fellowship Award, year of 2018-2019, Department of Physics and Astronomy, School of Arts and Sciences, University of Pennsylvania

2018 Outstanding Graduates of Beijing, Ministry of Education

2015/2016/2017 WeiMing XueZi Scholarship (CNY 17,000 Total), School of Physics, Peking University,

2016 National Scholarship of China (top 2%, CNY 8,000), Ministry of Education

2016 Outstanding Merit Student of Peking University

2013 Gold Medal in China Physics Olympics Final (rank 16/300)

Teachings: TA positions Held

Physics 151: Electromagnetism, TA, 2023 Spring

Physics 522: Introduction to Elementary Particle Physics, TA, 2023 Spring

Physics 632: Quantum Field Theory II, Grading, 2022 Spring

Physics 601: Introduction to Quantum Field Theory, Grading, 2021 Fall

Physics 151 Labs: Electromagnetism (Remote), Grading, 2021 Summer

Math 312: Linear Algebra, Grading, 2021 Spring

Math 114 Calculus II: Multivariable Calculus, Leading recitations + grading, 2020 Fall

Physics 101 Labs: Mechanics (Remote), Grading, 2020 Summer

Physics 151 Labs: Electromagnetism (Remote), Grading, 2020 Summer

Physics 151: Electromagnetism, Grading, 2020 Summer

Math 210: Math in the age of info, Grading, 2020 Spring

Math 104 Calculus I: Integrals and Series, Leading recitations + grading, 2019 Fall

Physics 151 Labs: Electromagnetism, Leading recitations + grading, 2019 Summer

Physics 171: Honor Electromagnetism, Grading, 2019 Spring

Physics 170: Honor Mechanics, Grading, 2018 Fall

Computational Physics (Peking University), Leading recitations + grading, 2018 Spring